MILITARY SPECIFICATION
PERISCOPE, T33

1. SCOPE

1.1 This specification covers one type of periscope known as Periscope, T33. This periscope is a component of Computer T27E2, which is part of Fire Control System T38.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and drawings and those specifications and drawings referenced thereon, of the issue in effect on the date of invitation for bids, form a part of this specification.

SPECIFICATIONS

Military	
MIL-F-13926	Fire Control Materiel; General Specification Governing the Manufacture, and Inspection of
MIL-O-13830	Optical Components for Fire Control Instruments; General Specification Governing the Manufacture, Assembly and Inspection of

DRAWINGS

Ordnance Corps	
K7629516	Periscope, T33

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 General.- Unless otherwise specified, the requirements of Specification MIL-O-13830 and Specification MIL-F-13926 shall be mandatory..

3.2 Material.- Material shall be in accordance with the drawings and specifications forming a part of this specification.

3.3 Electrical

3.3.1 High potential.- There shall be no puncture of insulation, no sparkover, or decrease in dielectric strength of the insulation sufficient to cause failure of the periscope when tested in accordance with 4.2.1

3.3.2 Insulation resistance.- All insulation shall have a minimum resistance value of 100 megohms, except when stipulated otherwise in detail Government specifications and drawings. Test shall be in accordance with 4.2.2.

3.4 Performance

3.4.1 Operational limits.- The periscope shall operate within the following limits:

Azimuth coverage	6400 mils continuous	
Elevation coverage (for center of field)	-170 to + 1600 mils	
Head servo (azimuth)		
Velocity	1250 mils/sec	minimum
Acceleration	1800 mils/sec^2	minimum

3.4.2 Operational check.- The heater and reticle light shall function properly. The cam switch spacing shall be adjusted to obtain minimum hunting without interference by the head window, except for some vignetting.

3.4.3 Shaft torque.- The maximum torque required to turn either A_o or E_o shafts at the input couplings shall be 2.0 in. oz. There shall be a negligible amount of interaction between the E_o and A_o couplings, such that a 0.5 in. oz. torque on one coupling (without universal shafts in place) shall show no movement when the other is rotated.

3.4.4 Eyepiece focus.- The focus of the eyepiece shall be between minus 0.75 and minus 1.0 diopter when focused on a target 200 to 250 yards distant. (Dioptometer standard).

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3.4.5 Definition.- Resolving power for a parallel bar test object shall be 25 seconds of arc or better. The test object shall be of the pattern illustrated in Specification MIL-O-13830. The image shall be viewed with the aid of a 6 power, 3.5 mm aperture auxiliary telescope of approved design, scaled in diopters. The test pattern shall be resolved in each of the four meridians by an adjustment of the auxiliary telescope not to exceed ± 1/8 diopters from the setting for best focus on the numeral in the center of the test field.

3.4.6 Parallax.- Parallax in the center of the field shall not exceed 0.25 mils when viewing a clearly defined target at a distance of 200 to 250 yards.

3.4.7 Parallelism between reticle and image.- With the mounting surface of the periscope vertical and the head rotating, the vertical line of the reticle shall remain parallel to the image of a plumb line within 0.5 mils over the entire length of the reticle pattern.

3.4.8 Line of sight control

3.4.8.1 Azimuth.- From zero azimuth at each increment of 13° 20' for one complete revolution in azimuth (controlled from the input coupling) the position of the line of sight shall be within 3' 22" of the true azimuth and elevation position. Each point shall be approached both from an increasing and from a decreasing value of azimuth. True position is defined as that point which makes the error value of azimuth a minimum.

3.4.8.2 Elevation.- From minus 6° elevation at each increment of 5° 20' until 90° is reached (controlled from the input coupling) the position of the line of sight shall be within 3' 22" of the true elevation and azimuth position. Each point shall be approached from both an increasing and from a decreasing value of elevation. True position is defined as that point which makes the error values of elevation a minimum.

3.4.9 Backlash.- The backlash in the line of sight position for either azimuth or elevation shall not exceed 1.0 mil at any measured point.

3.4.10 Image displacement.- The displacement of the reticle image in the eyepiece focal plane due to dove prism adjustment, pyramidal error, or optical misalignment shall not exceed ± 9.0 mils with respect to the test reticle at this focal plane.

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3.4.11 Alignment of objective reticle on target.- When the optical tube within the telescope assembly K7629655 is rotated, the shift of the target with respect to the reticle shall not exceed .25 mils when the objective is focused on a target at a distance of 200 to 250 yards.

3.4.12 Periscope head control.- The cams that control the head control servo motor shall be set so as to give minimum jitter when following the head prism. When the line of sight is at zero elevation and the azimuth input shaft is turned so as to rotate the line of sight at an acceleration of 1800 mils/sec^2, the edge of the periscope head window shall not reduce the field of vision of the periscope.

3.4.13 Image tilt.- The image of a plumb line formed by the periscope shall remain vertical within 30 minutes when the periscope head is traversed through two complete revolutions.

3.4.14 Vibration.- After being subjected to the vibration test of Specification MIL-0-13830 there shall be no damage to the instrument, no change in collimation in excess of 0.2 mil, and no loose dirt in excess of the following:

(1) A dirt particle on the reticle, or on the second surface of the objective lens #3 and the first surface of the first erector, shall not exceed one in number on each surface of a size not larger than the width of a reticle line thickness.

(2) Dirt on the second surface of the filter and the first surface of the eyepiece shall not exceed a total of four in number of a size (apparent) not larger than a number 20 dig.

(3) Dirt in the eyepiece shall not exceed three in number of a size (apparent) not larger than a number 20 dig.

(4) The maximum size dirt may be exceeded in length by twice the width size. Smaller dirt size may be proportionally more in number.

3.5 Environmental

3.5.1 Temperature.- The periscope shall not suffer damage or fail to meet the requirements of 3.4 when tested in accordance with 4.3.2.

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3.5.2 Humidity.- The periscope shall not suffer damage or fail to meet the requirements of 3.4 when tested in accordance with 4.3.3.

3.5.3 Rain.- There shall be no evidence of water penetration or damage to the periscope when tested in accordance with 4.3.4.

3.5.4 Sand and dust.- The periscope shall not suffer damage or fail to meet the requirements of 3.4 when tested in accordance with 4.3.5.

3.5.5 Shock.- The periscope shall not suffer damage or fail to meet the requirements of 3.4 when tested in accordance with 4.3.6.

3.6 Workmanship.- Workmanship shall be in accordance with Specification MIL-O-13830 and Specification MIL-F-13926.

4. QUALITY ASSURANCE PROVISIONS

4.1 General.- Unless otherwise specified, each periscope shall conform with the quality assurance provisions of this specification, Specification MIL-O-13830, and Specification MIL-F-13926.

4.2 Electrical tests

4.2.1 High potential.- The insulation between ungrounded circuits and ground, and between isolated circuits shall be subjected to a potential of 1,250 volts (rms) at a frequency of 60 cycles per second. The test voltage shall be applied instantaneously for a period not less than 5 seconds or more than 30 seconds. Insulation resistance test shall be performed immediately after this test.

4.2.2 Insulation resistance.- The insulation resistance between ungrounded circuits and ground, and between isolated circuits shall be measured by an approved method using a source of d-c potential of 500 volts.

4.3 Environmental tests

4.3.1 General.- One of the first five periscopes and one out of every additional fifty or fraction thereof shall be tested. A unit that has satisfactorily complied with the requirements of 3.4 shall be subjected to all of the environmental tests.

4.3.2 Temperature.- A periscope prepared for normal operation shall be placed in the chamber and exposed to an air temperature of minus 62°C (- 80°F) for a period of 3 days, without heaters operating. The air temperature then shall be raised to minus 40°C (- 40°F) and the heaters operated until the internal temperature of

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the unit has reached equilibrium, it shall be tested for conformance with 3.4.1, 3.4.3, 3.4.9 and 3.4.12. Remove all power from the unit and raise the air temperature to normal room temperature and test for conformance with 3.4. Raise the air temperature to plus 71°C (+ 160°F) and after the internal temperature of the unit has reached equilibrium at that temperature allow it to remain at that temperature for a period of 4 hours. Reduce the air temperature to plus 52°C (+ 125°F) and after the internal temperature of the unit has reached equilibrium at that temperature expose the top surface to 360 BTU/Ft. Sq/Hr., (50 to 60 percent of the total energy shall be in wave lengths above 8,000 angstrom units and 4.5 to 6.0 percent in wave lengths below 4,000 angstrom units) for a period of 4 hours. The unit then shall be tested for conformance with 3.4.1, 3.4.3, 3.4.9 and 3.4.12. Allow the unit to return to normal ambient temperature and test for conformance with 3.4.

4.3.3 Humidity.- A test chamber that is capable of being sealed shall be used, the volume of which shall exceed the total volume of the periscope being tested by 50 percent or more. Distilled or demineralized water having a pH value of between 6.5 and 7.5 at 25°C (77°F) shall be used to obtain the desired humidity. A unit prepared for normal operation, shall be placed in the chamber so that it is no closer than six inches from the sides of the chamber, and is not being subjected to radiant heat. The humidity of the chamber shall be adjusted to 95 ± 5 percent and maintained at that value during remainder of test unless noted otherwise. Raise the temperature of the chamber at a linear rate for a period of 30 minutes until a temperature of plus 46° ± 1°C (+ 115° ± 2° F) is reached, and maintain at that temperature for a period of 4.5 hours. Reduce the temperature of the chamber at a linear rate for a period of 1.0 hour until a temperature of plus 27° ± 1°C (+ 80° ± 2°F) is reached. The chamber temperature shall then be varied ± 5°C (± 9°F) at least once each hour for the next six hours; the rate of change shall be optional, consistent with maintenance of the maximum temperature not in excess of 20 minutes out of each hour. At least once each hour the relative humidity shall be 100% with condensation, the duration of which shall be optional and need not necessarily coincide with the maximum temperature phase of the hourly cycle. The above noted 12 hour cycle shall be repeated twenty times. During the low temperature period of the twentieth cycle, the unit shall be tested for conformance with 3.4.1, 3.4.3, 3.4.9 and 3.4.12.

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4.3.4 Rain.- A periscope prepared for normal operation shall be mounted in the test chamber. The rain test chamber temperature shall be maintained between 20°and 30°C (68°and 86°F) throughout the test period. A simulated rainfall of 4 ± 1 inch per hour (as measured at the surface of the unit with a U. S. Weather Bureau weighing-type gage) shall be produced by a suitable means of such design that the water is emitted in the form of small droplets rather than a fine mist. The temperature of the water shall be maintained between 11°and 24°C (52°and 75°F). The water spray shall be dispersed uniformly on the unit for a period of two hours in directions ranging from vertical to 45 degrees. The simulated rain shall be directed on all surfaces that are normally subjected to rain.

4.3.5 Sand and dust.- A periscope prepared for normal operation shall be mounted in the test chamber. The sand and dust density shall be raised and maintained at 0.1 to 0.5 grams per cubic foot within the test space. The relative humidity shall not exceed 30 percent at any time during the test. Sand and dust used in the test shall be of angular structure and shall have characteristics as follows:

(a) 100 percent of the sand and dust shall pass through a 100-mesh screen, U. S. Standard Sieve Series.

(b) 98 ± 2 percent of the sand and dust shall pass through a 140-mesh screen, U. S. Standard Sieve Series.

(c) 90 ± 2 percent of the sand and dust shall pass through a 200-mesh screen, U. S. Standard Sieve Series.

(d) 75 ± 2 percent of the sand and dust shall pass through a 325-mesh screen, U. S. Standard Sieve Series.

(e) Chemical analysis of the dust shall be as follows:

Substance	Percent by weight
SiO_2	97 to 99
Fe_2O_3	0 to 2
Al_2O_3	0 to 1
TiO_2	0 to 2
MgO	0 to 1
Ign losses	0 to 2

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The internal temperature of the test chamber shall be maintained at 25°C (77°F) for a period of 6 hours with sand and dust velocity through the test chamber between 200 ± 100 feet per minute. The temperature shall then be raised and maintained at 71°C (160°F) for 6 hours with sand and dust velocity as stated above. Allow the unit to return to normal ambient temperature and test for conformance with 3.4.

4.3.6 Shock.- The periscope shall be mounted rigidly, using its own mounting provisions, to simulate field use. A shock of 20 g's shall be imparted three times in each direction along each of the three mutually perpendicular directions, making a total of 18 shocks. Each shock impulse shall have a time duration of 11 ± 1 milliseconds; the maximum g value shall occur at approximately 5.5 milliseconds and shall be within ± 10 percent. Filters used shall have cut-off frequencies selected to eliminate all harmonics above the fundamental, from the measured shock impulse. At the completion of this test the periscope unit shall be tested for conformance with 3.4.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing and marking shall be in accordance with instructions supplied by the mission arsenal through the contracting officer.

6. NOTES

6.1 Intended use.- Periscope, T33 is designed for use with the Skysweeper as a means of tracking targets optically.

6.2 Ordering data.- Contracts and invitation for bids should specify the title, number, date of this specification and type of packaging and packing.

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